

MINI PROJECT REPORT

Student Performance Analysis

Excel Data Preprocessing &
Power BI Interactive Dashboard



1. Introduction

AI Impact on Student Performance is a data analytics project that studies student academic performance together with AI usage, attendance, grade level, AI tools, AI usage purpose and AI dependency. The project follows a two-phase workflow: Phase 1 focuses on understanding and cleaning the dataset, while Phase 2 converts the cleaned data into an interactive Power BI dashboard.

The purpose of the analysis is to understand patterns in student performance and explore how AI usage and related student behaviour are represented in the dataset. The final dashboard provides a structured way to view overall performance, AI usage patterns and student-level details.

2. Objectives

- Analyze overall student performance using final score and related measures.
- Study AI users and non-AI users and examine AI usage patterns.
- Analyze the distribution of AI tools and AI usage purposes.
- Examine relationships between AI usage, attendance, AI dependency and final score.
- Develop an interactive Power BI dashboard using KPIs, charts, slicers, navigation, bookmarks and drill-through.

3. Dataset Description

Dataset: AI Student Performance Dataset.

The Phase 1 report records 8,000 student records and 26 columns. Microsoft Excel was used for the initial data understanding and cleaning stage.

The cleaned dataset used in Phase 2 contains fields such as `student_id`, `grade_level`, `uses_ai`, `ai_tools_used`, `ai_usage_purpose`, `attendance_percentage`, `final_score` and `ai_dependency_score`.

3.1 Important Fields

- `student_id` – student identifier used for student-level analysis and drill-through.
- `grade_level` – student grade-level category.
- `uses_ai` – indicates AI usage status.
- `ai_tools_used` – records the AI tools used.
- `ai_usage_purpose` – records the purpose of AI usage.
- `attendance_percentage` – attendance measure.
- `final_score` – final academic performance score.
- `ai_dependency_score` – measure used for AI dependency analysis.

4. Tools & Technologies Used

- Microsoft Excel – dataset understanding, data cleaning and preparation.
- Microsoft Power BI – dashboard development and interactive analytics.
- Power Query – transformations may be used during data preparation where applicable; the supplied reports do not document specific Power Query steps.
- DAX – not specifically documented in the supplied Phase 1/Phase 2 reports, so no DAX measures are claimed here.

5. Data Cleaning & Transformation

Phase 1 documents the data-quality checks completed before visualization. Duplicate records were checked using Remove Duplicates and no duplicate rows were found. Blank or missing values were checked using Excel filters and no actual blank cells were identified in the checked dataset.

The existing value 'None' in categorical fields such as ai_tools_used and ai_usage_purpose was preserved as a dataset value. Numerical and decimal values were checked for appropriate formatting, age values were checked against the range used during cleaning, and categorical fields were checked for consistency.

5.1 Cleaning Checklist

- Duplicate records – checked; no duplicates found.
- Blank/missing cells – checked; no actual blanks identified.
- Categorical values – checked for consistency.
- Numerical/decimal values – checked for appropriate formatting.
- Cleaned Excel dataset – prepared for analysis.

6. Data Analysis & Dashboard Development

The cleaned data was converted into a three-page Power BI report. The report structure moves from an overall summary to AI-specific analysis and then to student-level detail.

6.1 Report Pages

- Overview Dashboard – KPI cards for Total Students, Average Final Score, Average Attendance and AI Users, together with charts and slicers.
- AI Usage Analysis – AI Tools Used Distribution, AI Usage Purpose Distribution, AI Dependency Score analysis and AI Usage Purpose vs Average Final Score.
- Student Performance – Student Details table and drill-through destination for selected students.

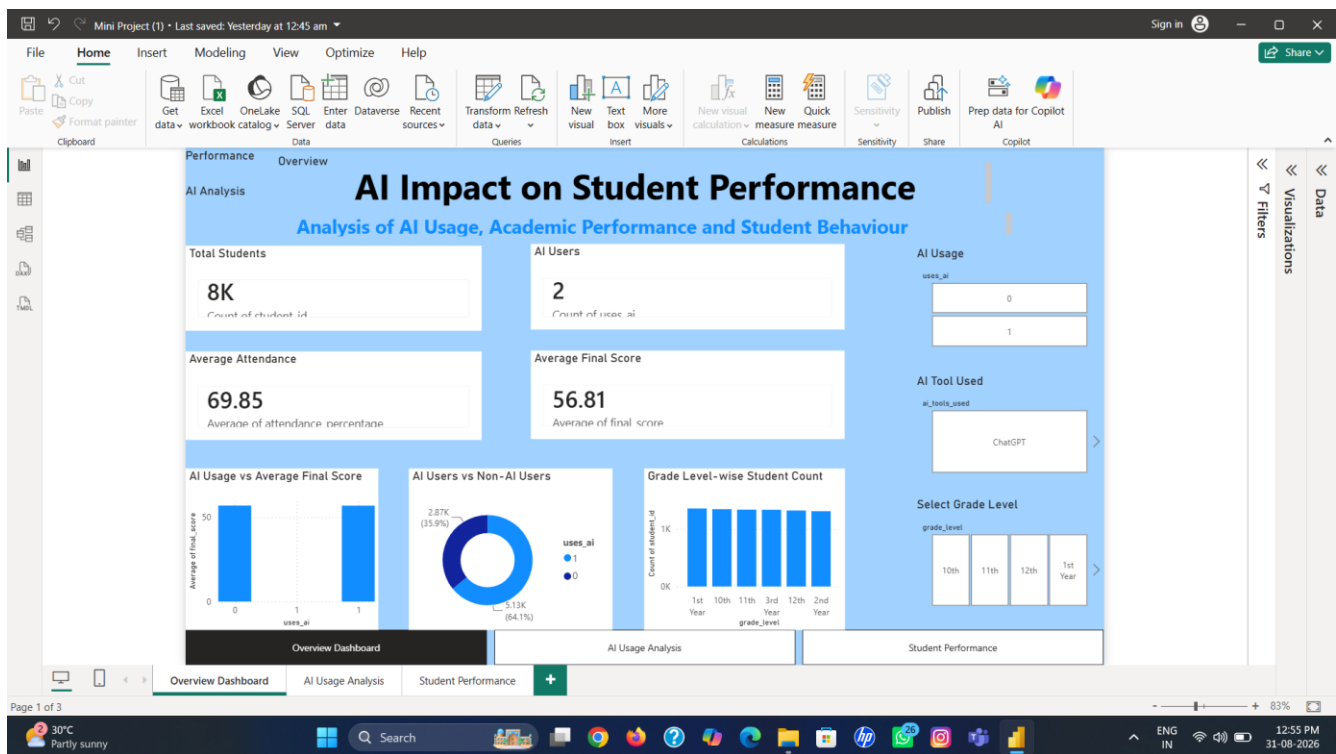
6.2 Interactive Features

- KPI cards for quick summary measures.
- Charts for comparing AI usage, grade level, attendance and final score.
- Slicers for dynamic filtering.
- Page navigation buttons between report pages.
- Bookmarks for predefined report views/states.
- Drill-through using student_id to open student-level details.
- Back button to return from the detail page.

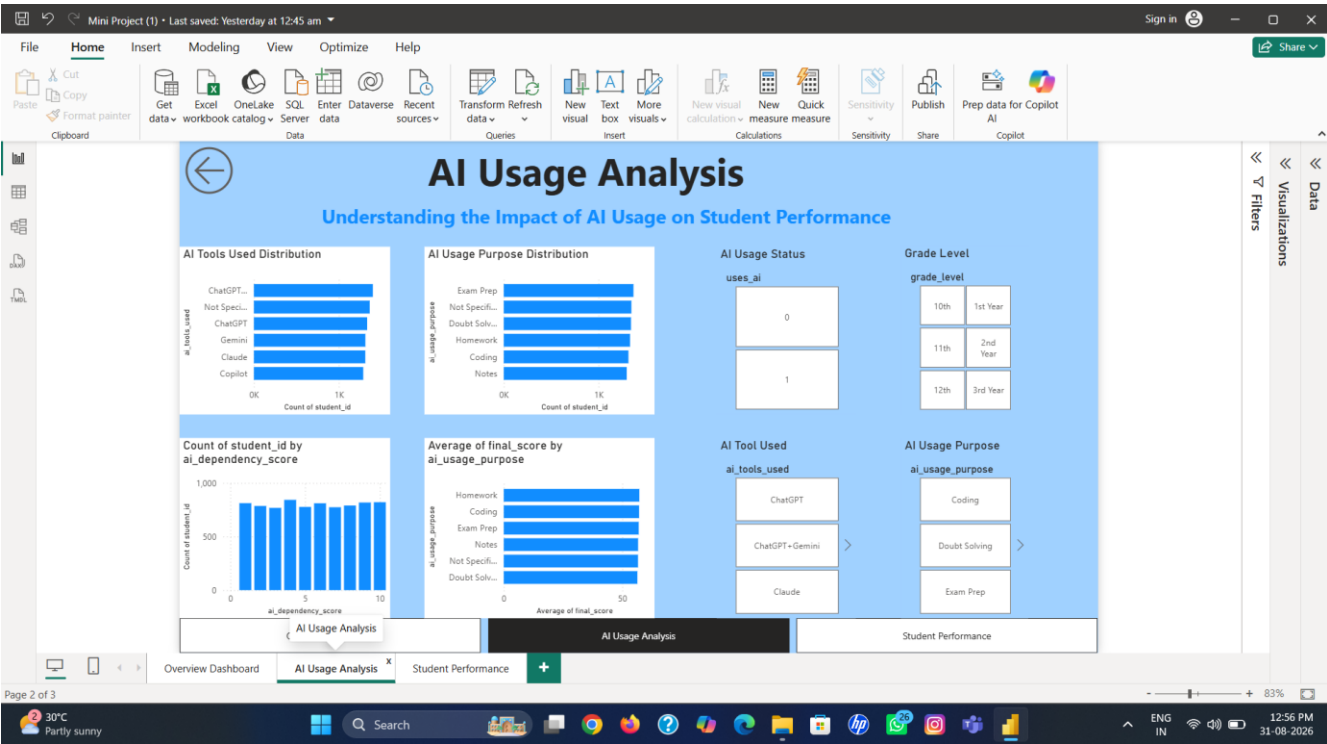
7. Dashboard Screenshots

Insert the final Power BI dashboard screenshots in this section before submission. The supplied Phase 2 documentation describes three report pages but does not contain the actual dashboard images.

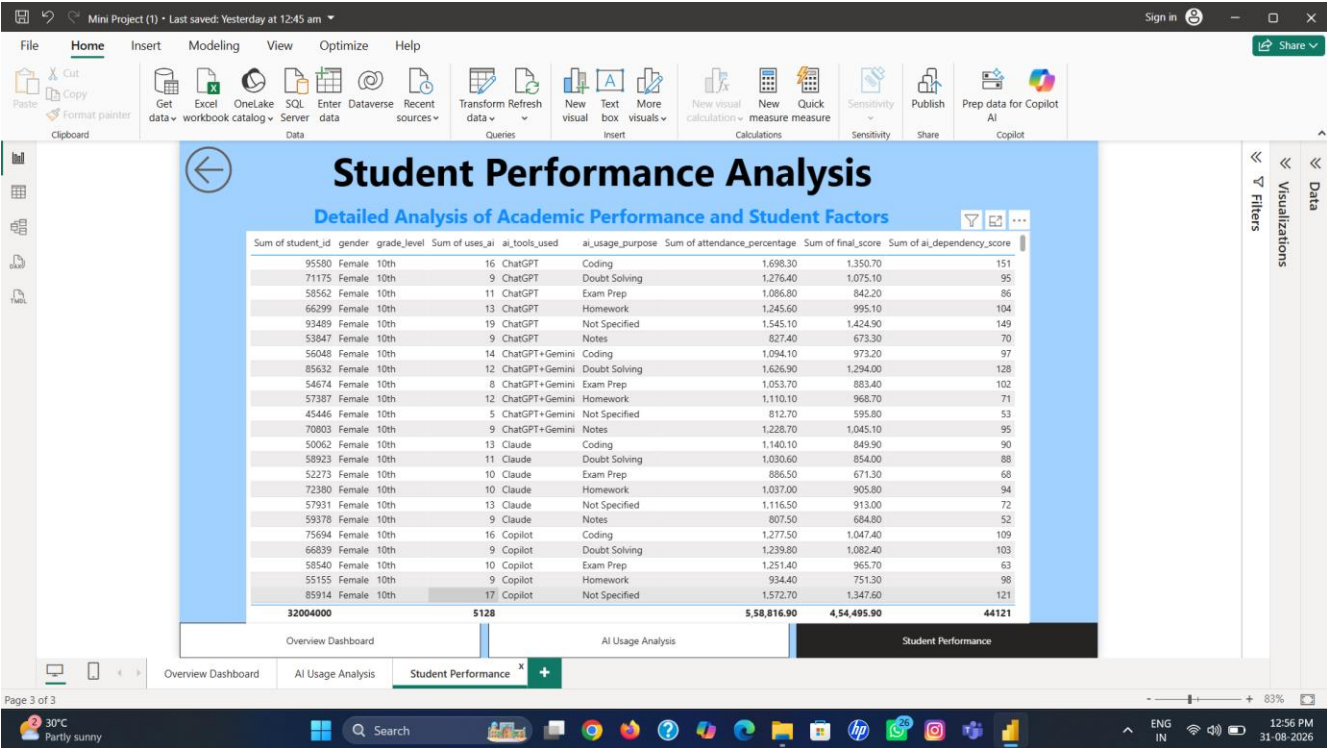
7.1 Full Overview Dashboard Screenshot



7.2 AI Usage Analysis Screenshot



7.3 Student Performance / Drill-through Screenshot



8. Key Insights / Findings

The supplied Phase 1 and Phase 2 reports document the analysis structure and the visuals to be used, but they do not provide actual numerical findings such as the highest/lowest sales, best-performing product, revenue, profit or sales trends. Because this project is about student performance rather than retail sales, those retail-specific findings are not applicable.

- The dashboard is structured to study student performance and AI usage.
- AI user distribution is presented on the Overview Dashboard.
- AI tools and AI usage purposes are analyzed on the AI Usage Analysis page.
- AI usage is compared with average final score.
- AI dependency score is included as an analysis dimension.
- Attendance percentage is included as a performance-related analysis dimension.
- Student-level investigation is supported through drill-through using student_id.

For the final submission, exact numerical findings should be added from the completed Power BI visuals after checking the actual dashboard values.

9. Conclusion

The project provides a complete workflow from dataset understanding and cleaning to interactive dashboard development. Phase 1 establishes a checked and cleaned dataset, while Phase 2 uses that dataset to create a three-page Power BI report for overall performance, AI-specific analysis and student-level investigation.

The dashboard improves usability through KPI cards, charts, slicers, navigation buttons, bookmarks and drill-through. Before final submission, the drill-through field should be verified as student_id and the complete right-click → Drill through → Student Performance workflow should be tested.

10. Future Scope

- Forecasting future student performance and AI-usage trends.
- Connecting the dashboard to real-time or regularly refreshed data.
- Automating scheduled reports and distribution.
- Adding more advanced statistical or predictive analysis.

11. References

- AI Student Performance Dataset – source information as provided with the project dataset.
- Phase 1 Report – AI Student Performance Analysis: Data Understanding & Data Cleaning Report.
- Phase 2 Report – AI Impact on Student Performance: Power BI Dashboard Development & Interactive Analytics.
- Microsoft Power BI documentation – general reference for dashboard and visualization concepts (specific web references were not listed in the supplied reports).